

A Logarithmic Lower Bound for Adversarial Delays

Theorem 1. *For every horizon $T \geq 2$ and every possibly randomized online policy π , there exists a deterministic, oblivious, nonincreasing price sequence*

$$1 = p_1 \geq p_2 \geq \dots \geq p_T > 0$$

such that

$$\mathcal{V}_T = 0$$

and

$$R_T \geq \frac{3}{16}(H_T - 1).$$

Proof. The proof uses a distribution over deterministic oblivious sequences. All random choices defining the sequence are sampled before the game starts. We prove that the regret averaged over this distribution is at least the desired quantity; hence at least one deterministic sequence in its support has at least that much regret.

For $t = 2, \dots, T$, set

$$\varepsilon_t := \frac{1}{2t}.$$

Let $Z_2, \dots, Z_T$ be independent Bernoulli random variables satisfying

$$\mathbb{P}(Z_t = 1) = \varepsilon_t.$$

Define the random price sequence $(P_t)_{t=1}^T$ by

$$P_1 := 1, \quad P_t := \prod_{s=2}^t (1 - \varepsilon_s)^{Z_s}, \quad t \geq 2.$$

Equivalently, conditional on $P_{t-1} = x$,

$$P_t = \begin{cases} x, & \text{with probability } 1 - \varepsilon_t, \\ (1 - \varepsilon_t)x, & \text{with probability } \varepsilon_t. \end{cases} \quad (1)$$

Every realization is nonincreasing. Therefore, for every realization $\mathbf{p}_T$,

$$M_t = p_t \quad \text{for every } t,$$

and consequently

$$\mathcal{V}_T = 0. \quad (2)$$

Lemma 1. *Fix $t \geq 2$. Conditional on any price history with $P_{t-1} = x$, the conditional expected reward of the learner at round t is at most*

$$(1 - \varepsilon_t)x.$$

Proof. Let $A_t = a$. There are three cases.

If $a < (1 - \varepsilon_t)x$, then the order executes in both possible states, and its reward is

$$a \leq (1 - \varepsilon_t)x.$$

If

$$(1 - \varepsilon_t)x \leq a < x,$$

then the order executes only when $P_t = P_{t-1}$, and its conditional expected reward is

$$(1 - \varepsilon_t)a \leq (1 - \varepsilon_t)x.$$

Finally, if $a \geq x$, the order never executes and its reward is zero. □

Let

$$m_t := \mathbb{E}[P_t], \quad t = 1, \dots, T,$$

and introduce the convenient conventions

$$m_0 := 1, \quad \varepsilon_1 := 0.$$

Lemma 2. *For every $t = 1, \dots, T$,*

$$m_t = (1 - \varepsilon_t^2)m_{t-1}. \tag{3}$$

Moreover,

$$m_t \geq \frac{3}{4} \quad \text{for every } t = 0, \dots, T. \tag{4}$$

Proof. The claim is immediate for $t = 1$. For $t \geq 2$, conditioning on $P_{t-1} = x$ and using (1) gives

$$\begin{aligned} \mathbb{E}[P_t \mid P_{t-1} = x] &= (1 - \varepsilon_t)x + \varepsilon_t(1 - \varepsilon_t)x \\ &= (1 - \varepsilon_t^2)x. \end{aligned}$$

Taking expectations proves (3).

Iterating the recursion gives

$$m_t = \prod_{s=2}^t \left(1 - \frac{1}{4s^2}\right).$$

For numbers $x_i \in [0, 1]$, one has $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$. Hence

$$\begin{aligned} m_t &\geq 1 - \sum_{s=2}^t \frac{1}{4s^2} \\ &\geq 1 - \frac{1}{4} \sum_{s=2}^{\infty} \frac{1}{s(s-1)} \\ &= 1 - \frac{1}{4} = \frac{3}{4}, \end{aligned}$$

which proves (4). □

Lemma 3. *For every realization $\mathbf{p}_T$ of the random sequence,*

$$B_T(\mathbf{p}_T) \geq T p_T.$$

Proof. Because $\mathbf{p}_T$ is nonincreasing, for every $a < p_T$ one has $a < M_t = p_t$ for every t . Therefore

$$\sum_{t=1}^T r_t(a) = Ta.$$

Taking the supremum over $a < p_T$ yields

$$B_T(\mathbf{p}_T) \geq \sup_{a < p_T} Ta = Tp_T.$$

□

We now combine the three lemmas. At round 1, the learner's reward is at most $1 = m_0$. For every $t \geq 2$, Lemma 1 gives

$$\mathbb{E}[r_t(A_t)] \leq (1 - \varepsilon_t)m_{t-1}.$$

Thus, with the convention $\varepsilon_1 = 0$, the learner's expected cumulative reward, where the expectation is now over both the random sequence and the learner's internal randomization, is at most

$$\sum_{t=1}^T (1 - \varepsilon_t)m_{t-1}. \quad (5)$$

By Lemma 3, the expected benchmark is at least Tm_T . Consequently, the regret averaged over the random price sequence satisfies

$$\mathbb{E}[R_T] \geq Tm_T - \sum_{t=1}^T (1 - \varepsilon_t)m_{t-1}. \quad (6)$$

We use the telescoping identity

$$\sum_{t=1}^T m_{t-1} - Tm_T = \sum_{t=1}^T t(m_{t-1} - m_t). \quad (7)$$

Indeed, expanding the right-hand side gives $m_0 + m_1 + \dots + m_{T-1} - Tm_T$. By (3),

$$m_{t-1} - m_t = \varepsilon_t^2 m_{t-1}. \quad (8)$$

Substituting (7) and (8) into (6) yields

$$\begin{aligned} \mathbb{E}[R_T] &\geq \sum_{t=1}^T (\varepsilon_t - t\varepsilon_t^2)m_{t-1} \\ &= \sum_{t=2}^T \left(\frac{1}{2t} - \frac{1}{4t} \right) m_{t-1} \\ &= \frac{1}{4} \sum_{t=2}^T \frac{m_{t-1}}{t}. \end{aligned} \quad (9)$$

Using Lemma 2,

$$\mathbb{E}[R_T] \geq \frac{1}{4} \cdot \frac{3}{4} \sum_{t=2}^T \frac{1}{t} = \frac{3}{16} (H_T - 1). \quad (10)$$

Therefore at least one deterministic sequence $\mathbf{p}_T$ in the support satisfies

$$R_T \geq \frac{3}{16} (H_T - 1).$$

This proves the theorem. □